

Forge and the commercial alternatives

Does the Next Gen Control Room LED wall need software we could have bought?

Prepared for the Forge exhibit · 24 August 2026 · Sources listed at the end. Claims are marked **VERIFIED** where a cited source states them and **UNVERIFIED** where they are inference or were not checked.

The short answer

Yes, products exist that do this, and they are mature. None of them is a drop-in for what this exhibit needs, and the cheap end of the market cannot do it at all. The single question that decides a buy is whether any vendor can put a **logged-in Honeywell dashboard** on the wall and keep it usable. No vendor documents that, so it can only be settled by a demo against a real dashboard. If a purchase is on the table, ask that first.

What the exhibit actually needs

Five requirements, in the order that they constrain the choice:

1. **Live, interactive web pages.** Not a video feed of a dashboard. An operator at the wall must be able to click and type, including through a single sign-on flow, with the session surviving.
2. **Several pages laid out on one wall**, each positioned and scaled independently, with no browser chrome.
3. **Rearrangeable on site**, against the real dashboards, at the real wall resolution, without editing config files at the venue.
4. **Controllable from a tablet or phone**, so an administrator is not tied to a keyboard at the wall.
5. **One wall, one PC.** A single 3840×2160 output. Not a multi-room, multi-display estate.

Requirement 1 eliminates most of the market. Requirement 5 makes most of the rest oversized.

The options

The enterprise control room tier

Four vendors sell software that genuinely does this. They are aimed at command centres and network operations rooms, and they are architected as *video wall controllers*: a server, or several, driving many displays.

PRODUCT	STRENGTH FOR THIS JOB	COST AND SHAPE	CONCERN
Useful	Web application sources, and operators can drive mouse and keyboard from a desktop or smartphone. The whole platform is managed from a browser or phone. VERIFIED	\$500 per display, per year. Subscription, so no capital approval. VERIFIED	Server based. Sized for estates far larger than one wall.
Polywall	A real shipping iOS app, plus Lens for remote wall access. Sources include URLs with dashboards and applications, and desktops with IP-KVM. VERIFIED	One licence covers every wall in a room. VERIFIED Pricing is not published. UNVERIFIED	Deploys as Server, Visualizer, Worker and Agent nodes. A large footprint for one PC.
Hiperwall	HiperSource Browser is a real Chromium browser that bypasses the screen capture approach entirely. HiperKVM adds keyboard and mouse control. VERIFIED	Licensed per display, and each browser tab consumes one HiperSource licence. VERIFIED	Cost scales with panel count, which is the thing you would want freedom to grow.
VuWall	Video wall management and KVM operations, strong AV-over-IP and security integrations. VERIFIED	Not published. UNVERIFIED	Oriented to AV infrastructure and security operations. More plumbing than this exhibit needs.

Barco Overture and Datapath were not evaluated. **UNVERIFIED**

The lightweight tier, and why it does not qualify

Free tools exist that look like an exact match and are not. The closest, GRID KIOSK, describes itself as a browser-based multi-pane website dashboard for kiosk use. It embeds pages with **iframes**, and its own documentation warns that sites may refuse to load because of X-Frame-Options or Content-Security-Policy. **VERIFIED**

That is fatal here, not inconvenient. Enterprise applications routinely forbid being framed, and single sign-on commonly breaks inside an iframe. This is the same reasoning that ruled iframes out of Forge's own design before any code was written. Multis, the other candidate, runs one browser per *monitor* rather than panels on one screen, which is a different problem. **VERIFIED**

How Forge compares

CAPABILITY	FORGE	COMMERCIAL TIER
Live interactive web pages, natively rendered	Yes	Yes, in all four
Handles an SSO login on the wall	Yes, measured	Not documented by any vendor
Several panels share one login	Yes	Not offered or documented
Rearrange the montage at the wall, on live pages	Yes	Layouts are authored in a console, not on the live wall
Named layout presets	Yes	Yes, the most consistently offered feature in the category
Control from a tablet or phone	Web page served by the wall	Yes, and Polywall ships a native iOS app
Health and status reporting	Per panel, over HTTP	Yes, and more mature
Redundancy and failover	No	Yes, via a second server
Camera, RTSP, HDMI and KVM sources	No, web only	Yes
Multi-display and multi-room estates	No, one output	Yes, this is what they are built for
Vendor support contract	No	Yes
Recurring licence cost	None	Per display, per year, and sometimes per browser tab

The honest reading of that table: **Forge is not filling a gap in the market.** It is narrower than any of these products, and it lacks redundancy, non-web sources and a support contract. What it has instead is an exact fit to a one-wall, web-only, SSO-protected exhibit, with two behaviours no vendor offers: arranging the montage on the live wall, and pointing several panels at one login.

Would buying have been cheaper?

On licence cost alone, yes. Useful at \$500 a year for one display is less than the engineering time already spent. The costs that do not appear on the invoice are a vendor dependency, a server deployment on the show PC, and an IT approval conversation about a third-party platform rather than an in-house application. On current evidence those are larger here than the licence, but that is a judgement, not a measurement.

What has actually been proven about Forge

Everything below was measured on the running application, not assumed. **VERIFIED**

- **Interactive panels need no coordinate translation.** Each panel is a native browser view, so input is routed and scaled to it by the browser engine, zoom included.
- **A login survives more than expected.** Cookies live in the session partition, not the renderer, so a session survives a reload, a fresh navigation, and even the view being destroyed and rebuilt. What a reload costs is the interaction in progress. This corrected a long-held assumption in the project's own design notes.
- **Memory is the real long-run risk.** Two panels and the overlay measured 699 MB, and 1414 MB was observed during testing. Panels can now be refreshed and their renderers recycled on a timer, never while in use.
- **Windows behaves.** The application's own end-to-end self-test passes on a Windows runner: panel management, mode handling, presets and the upkeep timers.
- **80 unit tests** cover configuration, layout geometry and the control server at close to full line coverage, plus 20 end-to-end assertions that gate every build.

What is not yet proven

One open risk outweighs the rest

Forge draws a transparent overlay above the live pages, and the whole architecture was chosen on the assumption that this composites correctly. That is **confirmed on macOS and unverified on Windows**, which is what the show PC runs. If it fails there, documented fallbacks exist, but they are more awkward. **Installing the current build on the show PC and looking at it is the highest-value hour available.** It is also the point at which a buy decision becomes much easier to argue.

- Builds are unsigned on both platforms, pending certificates.
- Auto-launch on boot and crash restart are not built.
- Behaviour over days rather than minutes, particularly memory, is untested.
- No vendor has been asked whether their product handles a real Honeywell SSO flow.

Recommendation

1. **Test the Windows overlay first.** It costs an hour and it either removes the largest technical risk or turns a buy into the obvious answer.
2. **Ask whether the venue already has a wall processor.** If a Useful, Barco or similar box is already specified for the build, its browser source may be free at the margin and vendor supported. That changes the calculus entirely and is a question for the AV integrator, not a procurement exercise.

3. **If buying, evaluate Useful first**, on published pricing and the documented combination of web sources with phone and tablet control. Take Polywall second, for the native iOS app, and get a quote early since pricing is private.
4. **Make an SSO demo the first vendor question**. Not a feature list. A real Honeywell dashboard, signed in, on their wall.
5. **Raise the optics**. This exhibit showcases Honeywell's own Next Gen Control Room. Useful, Hiperwall and VuWall all market themselves to control rooms. Running one inside Honeywell's control room story may be fine, or may be exactly wrong. That is a question for whoever owns the narrative.

Sources

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